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Bridging the Digital Divide: Empowering Rural Women Farmers Through Mobile Technology in Kerala

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Abstract: This research paper investigates the impact of mobile technology on empowering rural women farmers and promoting inclusive agricultural development. The primary emphasis is on how mobile technology enables rural women producers to become more self-reliant in agriculture, promotes market participation, strengthens social connections, facilitates socioeconomic integration, enhances rural quality of life, and fosters sustainable agricultural development. In addition, this study also analyses the influence of demographic factors on the use of mobile technology among rural women farmers. This study was carried out in Kerala, a renowned agricultural state in India. In Kerala, the Palakkad district is known to be the granary of Kerala due to the availability of fertile valleys, rivers, forests, and paddy fields. Data were collected from 192 rural women farmers from Palakkad via semi-structured interviews, questionnaire surveys, and focus group discussions (FGD). The data were analyzed using the paired *t*-test and Garratt ranking method. The results demonstrate that mobile technology markedly enhances access to agricultural information, market engagement, and social connectivity, resulting in greater empowerment for rural women farmers. Nonetheless, obstacles such as insufficient digital literacy and inadequate mobile infrastructure access persist as considerable impediments. The findings demonstrate the revolutionary potential of mobile phone use in bridging information gaps, empowering women farmers in rural areas, and advancing equitable agricultural development, as well as the barriers faced by rural women in using mobile technology.

Keywords: rural women farmers; mobile technology; AI; empowerment; financial inclusion; sustainable farming

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1. Introduction

Technological advancements have continuously changed industries, economies, and how people live and interact, from the Industrial Revolution to the Artificial Intelligence era (Xu et al., 2018) [1]. Mobile technology plays a prominent role in this transition. Smartphones, which have integrated mobile phones and personal digital assistant functions, play an important role in the way we work, communicate, and go about our daily lives (Chown & Nascimento, 2022) [2]. This technological revolution has profoundly influenced primary sectors, especially agriculture, by implementing innovations that improve efficiency, productivity, and sustainability (Brown et al., 2022) [3]. "Smart Farming" with "Smart Phones" is the recent trend. Mobile phones are indispensable communication instruments in different sectors, including agriculture (Khan et al., 2020) [4]. Agriculture has consistently prioritized communication and information. Since the early days of agriculture, livestock husbandry, and fishing, individuals have sought information from one another (Abu Rimi & Dr, 2017) [5]. The use of smartphones in agriculture is changing conventional farming practices into more accurate, data-driven strategies. In the agricultural sector, the digital technology era has led to the development of numerous software applications that are

easily accessible to producers and other stakeholders (Dehnen-Schmutz et al., 2016) [6]. The digital technology sector in India is currently at a critical juncture. It is imperative to promote a variety of digital initiatives, including precision farming, agricultural drones, mobile technology, e-Choupal, and Digital Green, on a large scale in order to enhance the adoption of new technologies by farmers. This will address issues such as low production, inconsistent product quality, inadequate access to diversified agricultural information, and a lack of knowledge about domestic and international markets (Mishra et al., 2024) [7].

Although the agricultural field has advanced technologically, the Global Report on Food Crises (GRFC) 2024 edition states that, in 2023, over 282 million people worldwide—a 24 million increase from the previous year—experienced high levels of acute hunger in 59 countries and territories. Therefore, governments and international organizations must give sustainable agriculture and rural development top priority, with a particular emphasis on improving the accessibility of resources, markets, and technologies for smallholder farmers. In many developing nations, women make up over 50% of the labor force employed in agricultural operations and are primarily accountable for ensuring household food security. As such, they are an essential component of farming households. Rural agricultural farming is significantly influenced by the contributions of women, who are the foundation of numerous farming communities. Empowering rural women farmers is central to sustainable development, as their leadership in agriculture and land stewardship is crucial for enhancing ecosystem health, ensuring food security, and promoting the long-term well-being of their communities (Akhter & Cheng, 2020) [8]. From the planting and care of crops to harvesting and the management of post-harvest activities, they are profoundly engaged in every aspect of agricultural production. In numerous regions, women are accountable for the preponderance of food production, which makes a substantial contribution to the local economy and food security. The majority of rural women play a critical role in sustaining their families by performing reproductive and productive tasks. Agricultural, livestock, vegetable, and handicraft production are the most significant activities among these diverse endeavors (Jabeen et al., 2020) [9]. The collective resilience and organizational strength of women producers within their communities are a prime example of their embodiment of "Power with", which allows them to collaborate, share resources, and achieve shared objectives (Gramm et al., 2020) [10].

Researchers and politicians have paid close attention to the issue of gender and agriculture in recent years, particularly in developing nations, where gender inequality is pervasive. Because many women experience gender-specific barriers that lower their productivity and limit their contributions to agricultural production, economic growth, and the well-being of their families and communities, there is a gender gap in the agricultural sector in many developing nations (Mulema & Damtew, 2016) [11]. Technology (ICT) has offered a variety of benefits to female farmers' livelihoods, improving their financial security and enabling the development of networks for knowledge sharing among farmers from comparable socioeconomic backgrounds (Rahman & Huq, 2023) [12]. Perceived utility and perceived simplicity of use are essential factors influencing user acceptance of information technology (Sullivan et al., 2022) [13]. The shift alters the use of modern technology and digital infrastructure to advance inclusivity and progress in the agriculture industry. The digitization of agriculture is becoming more attractive because of these conditions, as suggested by Kitole et al. (2024) [14]. Agricultural productivity and food security are significantly influenced by the agricultural practices of women producers in rural regions worldwide. Agricultural productivity and rural economies in India's rural heartlands are primarily reliant on the labor of women cultivators, who frequently encounter obstacles in terms of education, technology, and resources.

The introduction of mobile phones has significantly altered rural development, resulting in previously unimaginable opportunities for the advancement of women cultivators and inclusive agricultural growth. This study examines the impact of mobile technology on farmers' access to different agricultural operations. To assist farmers, various agricultural applications and programs are accessible, such as IFFCI Kisan, Kisan Suvidha, Krishify, eNAM, PM-Kissan, etc. In Kerala, women have long played a crucial part in farming operations. The traditional rural economy of the Palakkad district is characterized by agriculture and

other rural activities. Rural women producers are essential to the production and security of food (Butt et al., 2010) [15]. Paddy is the main crop in Palakkad. Over 40% of the state's paddy production comes from the Palakkad district, which is the state's largest producer. As a result, the value of the contribution provided by rural women farmers is acknowledged. This study gathered responses from 192 female rural farmers in Kerala's Palakkad district. Mobile phones have completely changed how rural women farmers interact with agriculture and given them much more power. Primarily, they have improved information accessibility, making it simple for them to obtain vital agricultural insights, such as market prices, crop cultivation techniques, weather forecasts, and pest management plans.

In addition, by avoiding conventional middlemen, mobile phones have given these women direct access to larger marketplaces, promoting economic prospects and market linkage. Furthermore, the accessibility of governmental programs and aid schemes is further improved by their availability through mobile platforms. Additionally, mobile technology has made it easier for rural farming communities to advance and become more sustainable by providing access to modern agricultural practices, such as enhanced crop varieties, precision farming techniques, mechanization, and systems for irrigation.

In light of ICT's capacity to advance gender equality and support sustainable rural development, mobile phone use plays a significant role in empowering rural women farmers and promoting inclusive agricultural development. By gaining insight into how mobile technology affects the lives of rural women farmers, stakeholders may create more efficient plans to capitalize on the technology's comprehensive potential for reducing poverty and promoting inclusive growth. This study aims to contribute to a future where mobile technology serves as a powerful tool for farmers, ultimately leading to improved food security, increased agricultural productivity, and sustainable rural development. As Kerala is an agro-based state in India, this study is of great importance to determine the current status of ICT in agriculture and developing policies.

The transformative potential of mobile technology in agriculture, particularly its capacity to improve producers' access to critical information, markets, and financial services, has been emphasized in a multitude of global studies. Even with this acknowledgment, there is still a significant void in research that is concentrated on the Palakkad district of Kerala, particularly with the demographic factor of age and its influence on the utilization of mobile technology among rural women farmers. It is essential to comprehend the manner in which various age groups of women farmers in Palakkad incorporate mobile technology into their agricultural practices in order to create targeted interventions that can significantly enhance their productivity and socio-economic empowerment. In the Palakkad district of Kerala, this study attempts to look into the transformative potential of mobile technology in agriculture, with a specific focus on its role in promoting sustainable agricultural development and empowering rural women farmers. Understanding the current status of information and communication technologies (ICTs) in agriculture is vital for the development of policy in the state, given its agro-based economy. Despite the fact that global studies have emphasized the advantages of mobile technology in enhancing access to information, markets, and financial services, there is a substantial gap in research that looks into the influence of demographic factors—particularly age—on the utilization of mobile technology by rural women farmers in Palakkad. This research aims to offer insights into the ways in which various age groups incorporate mobile technology into their agricultural practices through exploring the association between age and mobile phone use. In turn, it can help in the development of more effective and targeted interventions.

In addition, this research will assess the extent to which mobile phones facilitate the empowerment of rural women producers by improving their access to essential resources, information, and networks. It also intends to examine the constraints that rural women encounter when implementing mobile technology, including socio-economic factors such as caste, education, and economic status, which affect their capacity to completely capitalize on ICTs. This study aims to guide future policy interventions and support mechanisms that may enhance the adoption and positive impact of mobile technology, thereby ensuring

inclusive and equitable agricultural development for rural women producers, by addressing these research questions (Table 1).

Table 1. Performance indicator report of wireless subscribers in India.

S.No.	Date	Wireless Subscribers in India (in Million)	
		Urban	Rural
1	31 March 2024	634.47	531.02
2	31 February 2024	636.11	528.53
3	31 January 2024	633.96	526.75
4	31 December 2023	633.44	525.05
5	31 November 2023	630.72	523.45
6	31 October2023	630.36	520.62

Source: Indian Telecom Services Performance Indicator Report 2024.

The changes in wireless subscribers in urban and rural India from October 2023 to March 2024 are the focus of the Indian Telecom Services Performance Indicator Report. Urban areas had 634.47 million wireless subscribers as of 31 March 2024, while rural areas had 531.02 million subscribers. The number of urban subscribers decreased from 636.11 million in February 2024 to 634.47 million in March 2024, a minor decrease. Conversely, rural subscribers experienced an increase from 528.53 million to 531.02 million during the same period. This trend of increasing rural subscriptions and declining urban subscriptions is evident in the months that have preceded it. For example, urban subscribers decreased from 633.96 million in January 2024 to 633.44 million in December 2023, while rural subscribers increased from 526.75 million to 525.05 million. In the same vein, urban subscribers decreased from 630.72 million to 630.36 million from November 2023 to October 2023, while rural subscribers increased from 523.45 million to 520.62 million. Over the reported months, these figures suggest a consistent pattern of growth in rural wireless subscribers and a minor decrease in urban subscribers.

2. Literature Review

ICT's promise for rural development is acknowledged, but little is known about how it specifically affects Keralan women who work in agriculture. Studies that have already been conducted on the subject have not gone far enough in explaining how mobile technology might help to close the gender gap, increase access to agricultural resources, and encourage sustainable farming methods. By illustrating how mobile technology can improve farming productivity, increase market options, and promote socioeconomic empowerment for rural women in Kerala, this study seeks to close that gap (Gupta & Agrawal, 2024) [16].

Collaboration between researchers, industry, and policymakers has increasingly focused on creating ICT business models tailored to women. Many industry leaders now recognize the vital role gender plays in technological advancement. While uncertainty remains about translating this into profitable enterprises for women, there are promising examples of early adopters driving a more inclusive approach to ICT innovation, particularly in fields like agriculture and beyond (Larsson, 2012) [17].

The research on determinants influencing the implementation of information and communication technologies (ICTs) in agriculture reveals a notable deficiency in comprehending the appropriate utilization of ICT tools within rural farming communities. The implementation of ICT in agriculture in Rupandehi District, Nepal, is constrained by socio-economic issues such as farm size, gender, and income inequalities. This disparity underscores the necessity for focused governmental policies and the creation of ICT solutions that cater to the actual requirements of farmers. Further investigation into these elements can facilitate a reduction in the technological disparity in rural agriculture, hence improving sustainable farming methods (Singh & Aryal, 2023) [18].

The literature emphasizes that the strategic incorporation of ICT in Tripura's local administration goes beyond simple technological implementation; its objective is to empower

women by equipping them with the necessary tools, abilities, and platforms for significant participation. These ICT initiatives encompass digital literacy programs, interactive web platforms, capacity-building endeavors, and specialized communication channels. These initiatives are collectively intended to address current disparities, change the balance of power, and promote inclusive systems of government (Debbarma & Sharma, 2024) [19].

The studies reveal valuable insights into the difficulties encountered by the agricultural community in obtaining access to efficient technical education. It emphasizes the significant influence of the triple divide—digital, rural, and gender—especially on women farmers. The primary limitations found encompass socio-economic obstacles, unsupportive family members, financial hardships, digital illiteracy, lack of motivation to acquire new skills, and insufficient extended advising services. These obstacles cumulatively impede the technological empowerment of farmers, with women farmers being particularly impacted (Saha et al., 2024) [20]. Empowerment is a process that allows individuals to assert control over their lives and improve their engagement in society (Rappaport, 1981) [21].

The study undertaken in the Upper Brahmaputra Valley Zone of Assam identifies a research gap concerning the insufficient awareness and utilization of information and communication technologies (ICTs) among rural farmers, especially women. This study indicates that, although ICT tools can greatly improve agricultural output and market access, many farmers, particularly women, lack the requisite skills and expertise to utilize this technology successfully. Moreover, obstacles such as insufficient infrastructure, elevated costs, and a scarcity of pertinent material in local languages impede the widespread implementation of ICT in agriculture (Kalita & Deka, 2024) [22].

The literature study demonstrates that technology has played a crucial role in enabling farmers in cooperatives across South India to sustain their efforts in generating short films for inspection processes during the epidemic. This practice persisted even after the lifting of COVID-19 restrictions. In addition, the use of personal Facebook profiles by farmers for promotion and marketing serves as an essential type of unpaid work that is vital for the long-term viability of cooperatives.

In the context of market liberalization and globalization, there is a growing need for quick access to reliable information on crop possibilities, technology, inputs, production techniques, services, and market trends. Indian farmers have enthusiastically adopted a range of information and communication technology (ICT) models, programs, and initiatives, which have improved their agricultural practices and increased their participation in the global market. To demonstrate this occurrence, this study utilizes the data envelopment analysis (DEA) approach to evaluate the efficiency of the top 17 states in India. This research highlights the incorporation of tribal and rural populations into the digital era in a rapidly changing sociocultural environment (Mishra et al., 2024) [7].

The literature study clearly showed that ICT (information and communication technology) has a crucial impact on improving crop management, establishing market links, enabling financial access, and increasing social inclusion in urban agriculture (Gramm et al., 2020) [10]. ICT breakthroughs can transform crop management by providing up-to-date weather updates, soil analysis tools, and pest control applications. This can lead to increased production and decreased losses for urban farmers. In addition, ICT enables market transactions through online platforms and mobile applications, enhancing prospects for generating revenue. Moreover, the presence of mobile banking services and crowdfunding platforms simplifies financial accessibility for farmers, hence promoting investment in resources and improvements in production (R & S, 2024) [23].

A study found that young Vietnamese male farmers with better earnings, residing in remote areas away from local markets, and engaging in training programs, are more likely to embrace information and communication technology (ICT) tools, such as mobile phones, to market their produce. To successfully encourage farmers to utilize ICT tools for agricultural marketing, it is essential to take into account their demographic, socio-economic, situational, and institutional attributes. These variables should also inform the formulation of marketing information strategies aimed at small-scale farmers in emerging nations (Hoang, 2020) [24].

A previous study presents a comprehensive examination of agricultural entrepreneurship from the viewpoint of farmers, including policy suggestions for developing nations to bolster agricultural entrepreneurship by improving digital competencies. The integration of network information technology has had a profound impact on several aspects of life, including information access, interaction, and loan acquisition. This, in turn, has greatly influenced the entrepreneurial attitudes and actions of farmers (Cheng et al., 2024) [25].

A previous study examined the influence of mobile phones on agricultural production in Benin, utilizing data from 209 maize farmers in six departments. The findings suggest that mobile phones enable extension professionals to provide agricultural advice from a distance, hence minimizing the necessity for in-person visits. Hence, national leaders must prioritize the enhancement of telecommunications infrastructure to facilitate greater access to services that enhance the productivity of both individuals and the nation as a whole (Aminou et al., 2018) [26].

A study performed in Huong Son District, Vietnam, revealed a significant deficiency in the utilization of Information Communication Technologies (ICTs), especially mobile phones, for agricultural marketing. Notwithstanding the capacity of ICTs to empower farmers, particularly women, by delivering timely market information and diminishing transaction costs, the study revealed that adoption among female farmers remains minimal. The results highlight that socio-economic and demographic characteristics, including gender, income, and access to training, substantially affect the adoption of ICT tools. Women farmers, frequently deprived of access to these resources, are less inclined to benefit from ICT tools, hence constraining their engagement in contemporary agricultural markets. Targeting this deficiency through gender-specific training and resources can enhance the inclusion of women in agricultural development (Houghton, 2009) [27].

Although the existing literature recognizes the transformative potential of ICT in rural development, especially for women in agriculture, there are still gaps concerning the specific effects on women farmers in Kerala. Prior research has predominantly concentrated on overarching themes, such as access to agricultural resources and market involvement, frequently neglecting the intricate impacts of demographic characteristics, such as age, on mobile technology usage (Gupta & Agrawal, 2024; Singh & Aryal, 2023) [16,18]. This study seeks to address these gaps by investigating the effects of age on mobile phone usage and its resultant effect on empowerment among rural women in Kerala. In contrast to previous studies that have predominantly overlooked the intersectionality of gender, age, and technology adoption (Kalita & Deka, 2024) [22], our analysis utilizes rigorous methodologies, such as paired *t*-tests, the chi-square test, and Garrett ranking methods, to offer an in-depth understanding of the obstacles encountered by these women. We provide a concentrated contribution that complements existing research and guides policymakers in formulating targeted actions for better results in rural areas.

Research Questions

1. To examine the association between demographic factors such as age and the use of mobile phones.
2. To assess how mobile phones influence rural women farmers' empowerment.
3. To analyze the limitations of using mobile phones that rural women farmers experience.

Hypotheses

HO1. *There is no significant difference between rural women farmers' age and their use of mobile phones.*

HO2. *There is no significant difference between pre- and post-mobile phone usage and the empowerment of rural women farmers.*

3. Data and Methodology

This research employed primary data to gather data with the assistance of a questionnaire. The data were gathered through semi-structured interviews, questionnaire surveys, and focus group discussions (FGDs).

3.1. Description of the Study Area

Palakkad district, situated in Kerala, India, was the site of this investigation. Due to its strategic location, it is frequently referred to as the "Gateway of Kerala" and is located in the central region of Kerala. Palakkad is located between 76°02′ and 76°54′ east longitude and 10°20′ and 11°14′ north latitude. Palakkad district is divided into 94 panchayats and 13 divisions, which are a combination of semi-urban and rural areas. Palakkad serves as the district's administrative hub. The district has a total population of approximately 2.8 million individuals, with a nearly equal distribution of males and women. Palakkad boasts a substantial agricultural foundation, with a total cultivable area of approximately 140,000 hectares. Rice is the staple food, and vegetables, pulses, and coconut are the primary commodities that are cultivated. Additionally, the district is recognized for its cultivation of cash commodities, including spices and rubber. In Palakkad, a significant proportion of the population is involved in mixed farming, which involves the cultivation of crops and the rearing of livestock. The agricultural practices in the area are diverse. The district is distinguished by a tropical monsoon climate, which is characterized by substantial rainfall during the southwest monsoon season, which typically spans from June to September. Palakkad's landform is characterized by a combination of plains and hilly regions, which contribute to the district's diverse microclimates. Palakkad is a critical agricultural center in Kerala due to the diverse climatic and geographical conditions that facilitate a diverse array of agricultural activities (Figure 1).

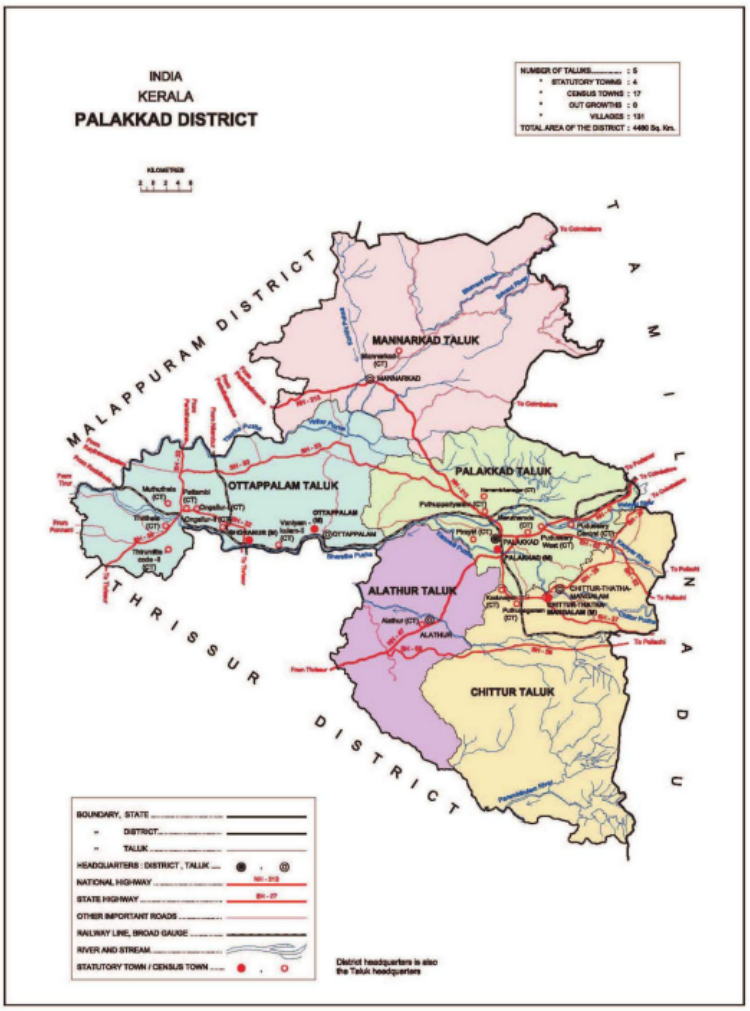


Figure 1. Map of the study area (Palakkad district). Source: District Resource Map, Palakkad District, Geological Survey of India (https://dmg.kerala.gov.in/wp-content/uploads/2020/01/dsr_pal.pdf, accessed on 9 August 2024).

3.2. Research Methodology

A mixed-methods approach was employed to acquire the data for this study. Semi-structured interviews were performed in conjunction with self-administered questionnaires. Furthermore, three focus groups, each including five members, were convened to obtain qualitative views. In order to comprehensively document the experiences, attitudes, and beliefs of rural women farmers in Palakkad with respect to the adoption and impact of mobile technology in agriculture, this methodology was selected. The women farmers' perceptions of the improvement in their agricultural practices prior to and following the implementation of mobile technology were evaluated through a self-administered questionnaire that included a combination of open-ended and closed-ended questions, as well as a four-point nominal scale.

Furthermore, a seven-point Likert scale was implemented to evaluate the obstacles and complications encountered by these producers when employing mobile phones in rural regions. The objective of this methodological approach was to facilitate the identification of barriers in using mobile technology. Consequently, a questionnaire was initially disseminated to 230 farmers, and 203 of them responded after persistent persuasion. Incomplete information resulted in the discarding of 11 out of 203 responses. To mitigate the potential for inadvertent bias, the data sources were triangulated through in-depth interviews with 192 participants in person. The questionnaires were translated and administered in the Malayalam language. The duration of each interview ranged from 35 to 45 min.

3.3. Tool of Analysis

In this case, the data were analyzed and interpreted using the Garrett ranking test to identify barriers to mobile technology use in rural areas and the paired *t*-test procedure to compare the improvement in respondents' agricultural practices before and after the implementation of mobile technology.

4. Data Analysis

Respondents were categorized based on their reported level of mobile device utilization when examining agricultural information, with categories including low, medium, and high usage. This measurement provides an understanding of the effectiveness with which rural women producers utilize mobile technology for agricultural purposes. A Chi-square test was performed on the number of respondents utilizing mobile devices to access agricultural information concerning age (Tables 2 and 3).

Table 2. Demographic data concerning age.

Age	Less than 25	26–40	41–55	More than 55
No. of respondents	13	53	88	38

Table 3. The number of respondents utilizing mobile devices to access agricultural information concerning age.

Age Group	No. of Respondents Using Mobile Phones to Look for Information About Farming.		
	Low	Medium	High
Less than 25	0	0	13
26–40	3	18	32
41–55	8	26	54
More than 55	8	28	2

The Chi-square test is a statistical technique that is employed to investigate the relationship between categorical variables. The statistical significance of the differences between the observed and expected frequencies is determined by tests. Here, a Chi-square test of

independence was executed to investigate the association between age group and mobile phone usage in the context of agricultural information. The data were categorized into four age groups and three levels of usage (low, medium, and high). The observed frequencies were entered into a contingency table, and the expected frequencies were determined using the underlying assumption that there is no relationship between the variables. The Chi-square statistic was calculated by contrasting the observed and expected frequencies (Table 4).

Table 4. Chi-square test.

O	E	O − E	(O − E)2	(O − E)2/E
0	1.29	−1.29	1.66	1.29
3	5.24	−2.24	5.02	0.96
8	8.71	−0.71	0.50	0.06
8	3.76	4.24	17.98	4.78
0	4.88	−4.88	23.81	4.88
18	19.88	−1.88	3.53	0.18
26	33	−7	49.00	1.48
28	14.25	13.75	189.06	13.27
13	6.84	6.16	37.95	5.55
32	27.88	4.12	16.97	0.61
54	46.29	7.71	59.44	1.28
2	19.99	−17.99	323.64	16.19

A Chi-square test of independence was performed to examine the relationship between age group and mobile phone usage in the context of agricultural information dissemination. The data were arranged in a contingency table, and the observed frequencies were compared with the expected frequencies calculated under the assumption of no relationship between the variables. O (Observed Frequency): The precise number of respondents in each age group and mobile usage category. E (Expected Frequency): the expected number of respondents in each category if there were no correlations between age and mobile phone usage. $(O - E)^2/E$: This value is used to determine the discrepancy between the expected and observed frequencies. The likelihood of there being a relationship between the variables increases as the discrepancy increases (Table 5).

Table 5. Result of chi-square test.

Chi-Square Value ($\chi 2 = \sum(Oi - Ei)2/Ei$)	Degrees of Freedom $(4 - 1) \times (3 - 1)$	Significance Level	Tabular Value	*p* Value
50.53	6	0.05	12.59	5.0302

Hence, if the Chi-square value is greater than the table value, the null hypothesis (HO1) is rejected. This would suggest that the levels of mobile phone usage for farming information vary significantly among various age groups. Age was found to be a key predictor of mobile phone use, with younger adults being more likely to adopt new technologies.

t-test on level of access to agricultural activities before and after mobile usage.

The t-test is used to determine if there is a significant difference between the means of two groups. The data illustrate the extent of access to a variety of agricultural activities both before and following the implementation of mobile phones. The data are presented in percentages for each access level (No, Low, Medium, and High) (Tables 6 and 7).

Table 6. Access percentage of various agricultural activities after the adoption of mobile phones.

Sl.No.	Activities	Level of Access to Activities in %							
		Before Mobile Usage				After Mobile Usage			
		No	Low	Med.	High	No	Low	Med.	High
1	Getting price information of agricultural products and inputs.	49	65	52	26	4	27	67	94
2	Access to the market to buy and sell products.	40	71	50	31	9	56	71	56
3	Voicing your opinion and suggestions about farming.	55	75	37	25	4	27	60	101
4	Attending classes relating to new cultivation methods, usage of new types of machinery, etc.	62	75	33	22	5	41	66	80
5	Access to finance (loan, insurance, and payments)	43	57	55	37	8	43	67	74
6	Connecting with community groups and other farmers to extent mutual support and knowledge exchange.	48	59	53	32	4	27	71	90
7	Updates regarding govt. schemes and services	57	69	40	26	6	29	66	91
8	Online purchase of agricultural machinery, pesticides, and other farm inputs.	87	59	24	22	6	27	70	89
9	Information access without physical travel	63	63	40	26	4	25	55	108

Table 7. Analysis of t-test.

	No	Low	Medium	High
t-values	10.27	8.64	−6.51	−9.84
Critical t-values	2.306	2.306	2.306	2.306
Degrees of freedom ($n-1$)	8			

At a 95% confidence level, the critical t-value of 2.306 is typically employed for a t-distribution with specific degrees of freedom. While comparing the absolute values of the calculated t-values with 2.306, it is evident that 10.27, 8.64, −6.51, and −9.84 were all greater than 2.306. As a result, the null hypothesis was rejected for all groups, as all of the calculated t-values were statistically significant, which suggested that there were significant disparities in access levels. This affirms that mobile phone usage has had a beneficial effect on producers' access to agricultural activities. In most activities, the comparison between pre-mobile and post-mobile usage reveals substantial improvements in access. The survey discovered that mobile phone users were capable of communicating with their relatives and migrant family members, as well as gathering information for agricultural and non-agricultural purposes (Mehta, 2016) [28]. Here, the data also suggest that mobile usage has significantly improved farmers' access to information, market participation, education, and overall information retrieval.

The development of women in rural areas can be considerably accelerated by the increasing availability of mobile phones at a low cost, which provides them with access to critical information on healthcare, agriculture, financing, and education. This connectivity enables women to make informed decisions, participate in entrepreneurship, and enhance their livelihoods, thereby promoting greater gender equality and community development. Agricultural input sources, contextual details such as location-specific weather forecasts and

market data, and the most effective agricultural practices are all areas in which farmers are continually seeking a variety of information to improve their farm business management. This information exchange is facilitated by the increasing adoption of mobile phones, which allows producers to access real-time updates on prices, demand indicators, and logistics, which are essential for making informed decisions.

Garrett Ranking Test

The Garrett Ranking Test is employed to evaluate and prioritize variables according to respondents' preferences. The method employed is to investigate the challenges associated with the use of mobile phones in rural areas. This analysis identifies the factor that has the most detrimental impact on the use of mobile phones in rural areas, enabling the ranking of the factors and the extraction of valuable insights. A scale was used to evaluate the impact of each issue, with higher scores indicating more significant challenges. The mean score for each problem was calculated by summing the scores assigned by the respondents to each problem. The problems were subsequently ranked based on their mean scores, with the highest score indicating the most urgent issue experienced by mobile phone users in rural areas (Table 8).

Table 8. Garrett ranking test on challenges associated with the use of mobile phones in rural areas.

S.No.	Problems with Using Mobile Phones in Rural Areas	1	2	3	4	5	6	7	No. of Respondents	Score	Mean Score	Rank
1	Dependency on male family members for monetary choices.	72	42	26	26	26	0	0	192			
		504	252	130	78	52	0	0		1016	112.89	3
2	A lack of education makes it difficult to use.	16	19	16	16	22	36	67	192			
		112	114	80	48	44	72	67		537	59.66	6
3	Limited coverage in rural regions.	81	56	29	26	0	0	0	192			
		567	336	145	78	0	0	0		1126	125.11	1
4	Lack of guidance on mobile phone usage.	16	16	19	19	22	35	65	192			
		112	96	95	57	44	70	65		539	59.89	5
5	Technical support services are lacking in rural locations.	70	47	28	24	23	0	0	192			
		490	282	140	72	46	0	0		1030	114.44	2
6	Mobile services available in regional languages.	25	41	33	0	0	45	48	192			
		172	246	165	0	0	90	48		721	80.11	4
7	Health risks from mobile phone radiation.	0	0	0	0	38	54	100	192			
		0	0	0	0	76	108	100		184	20.44	7

The Garrett Ranking of issues associated with mobile phone usage in rural areas indicates that limited coverage in rural regions is the most pressing issue, with a mean Garrett score of 125.11. The mobile network coverage and broadband connectivity have been identified as a concern in rural areas (Riddlesden & Singleton, 2014) [29]. The case is also the same here; the connectivity issues result in preventing rural members from accessing information. In the subsequent sections, dependency on male family members for monetary choices and technical support services are lacking, respectively, and have scores of 112.89 and 114.44. These scores underscore substantial concerns regarding financial autonomy and technical assistance. The lack of education making it difficult to use and lack of guidance on mobile phone usage are ranked fourth and fifth, respectively, suggesting substantial, but lesser, concerns. Health risks from mobile phone radiation are rated lower, with a mean score of 20.44, while mobile services available in regional languages are ranked sixth. In general, the rankings indicate that the most significant obstacles to mobile phone usage in rural areas are coverage and support issues, while language and health concerns are ranked lower.

5. Discussion and Conclusions

To enhance the adoption and effective utilization of ICT among rural women farmers, it is advised that they be provided with frequent, innovative ICT training sessions that are specifically tailored to their requirements. Rural communities should be provided with hands-on demonstrations and applications to familiarize them with their practical applications. The relevance of ICT in agricultural activities is generally regarded favorably by producers in the southern region of India (D. & Natarajan, 2020) [30]. Providing peer learning and mentorship opportunities with experienced users is essential to actively encourage aged rural women farmers to participate in these programs and to improve their skills. The potential for a significantly greater impact on rural productivity in the future is there as mobile penetration among agricultural communities continues to increase and information services continue to adapt and proliferate (Mittal et al., 2010) [31]. Farmers were using a significant amount of money to use transportation to reach the market to obtain the information and price of their products. However, they can obtain the most recent information on the spot by making a single call using their mobile phones. It is imperative to enhance the productivity of subsistence agriculture by improving the information, communication, business, and Internet resources available to farmers and the markets, organizations, and institutions with which they interact (Chhachhar et al., 2016) [32]. The implications of this study's findings align with the Mukhya Mantri Mahila Kisan Sashaktikaran Yojana, Agricultural Skill Development Training Programme for Women, and SHE Scheme by highlighting the need for targeted financial support, skill development, and capacity building for women in technical and agricultural sectors. This not only reinforces gender equality policies, like the Gender Equality and Women Empowerment Policy, but also stresses the importance of managerial practices that prioritize inclusivity, such as integrating women into leadership roles. Additionally, training programs like REACH emphasize bridging the skill gap, which can significantly enhance women's employability and career advancement.

The provision of technical assistance in rural areas and the resolution of mobile network issues necessitate government initiatives. The formulation of policies should be gender-sensitive, with a particular emphasis on the distinctive obstacles that women encounter in rural areas. To guarantee the long-term benefits and empowerment of rural women farmers, it is imperative to collaborate with local educational institutions or organizations to further integrate mobile technology into agricultural practices. The methodological part delineates the study techniques, although it is crucial to recognize certain constraints. This study's concentration on the Palakkad district in Kerala may restrict the applicability of its findings to other places with varying farming methods and levels of technological adoption. Moreover, disparities in digital literacy and access to mobile infrastructure among participants may affect the results, underscoring the necessity for an additional study to investigate these variables in a wider context.

In addition, future research should assess the efficacy of policy interventions and community support mechanisms in enhancing the adoption and impact of mobile technology among rural women producers. Mobile phone usage has a transformative impact on empowering rural women farmers by providing them with access to essential resources and information, thereby contributing to their participation in and contribution to agricultural development. Future research could investigate the reasons behind the limited impact on community groups and delve deeper into the factors influencing the significant rise in mobile usage for specific activities. The use of mobile phones by farmers for informational purposes should be proactively promoted by the government, non-governmental organizations, and other relevant authorities. Providing farmers with short-term, low-interest loans to help them purchase mobile phones is one way to encourage them. With the use of sophisticated software, seminars, and online training, farmers should be encouraged to use the latest innovations. These initiatives can enable farmers to recognize and resolve production and marketing-related issues.

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